How to use the DALI Language and Framework

EDIT AND RUN LOGIC MULTI-AGENT SYSTEM WITH **DALI**

OVER SICSTUS PROLOG

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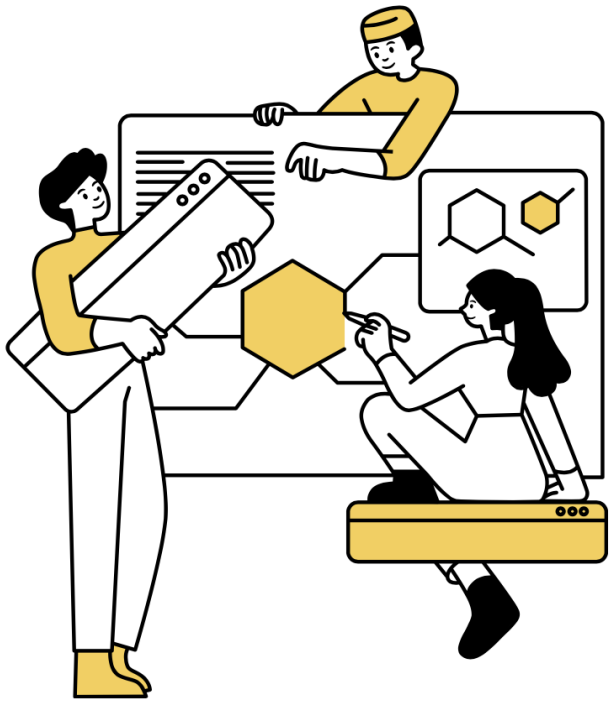
What is DALI ? Many concepts…



- A logic programming language
- A multi-agent development framework
- An extension of Prolog
- A Prolog meta-interpreter
- An event-based distributed software system with a shared message broker (Linda)
- A back-end micro-service tool with reasoning
- IT DOES NOT PROVIDE A GRAPHICAL USER INTERFACE

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But first…
Multi-Agent Software Engineering

Wooldridge, M., Jennings, N. R., & Kinny, D. (2000).

The Gaia methodology for agent-oriented analysis and design. *Autonomous Agents and multi-agent systems*, *3*(3), 285-312.

https://link.springer.com/article/10.1023/A:1010071910869

https://eprints.soton.ac.uk/253748/1/jaamas2000.pdf

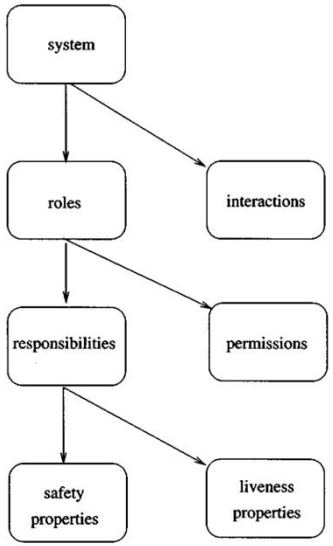
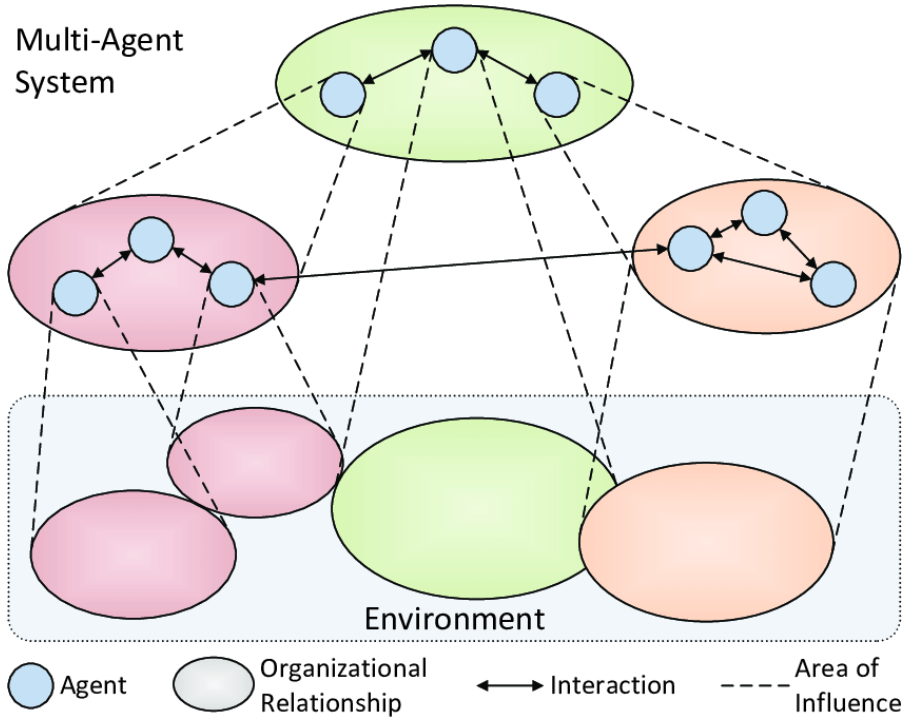


Table 1. Abstract and concrete concepts in Gaia

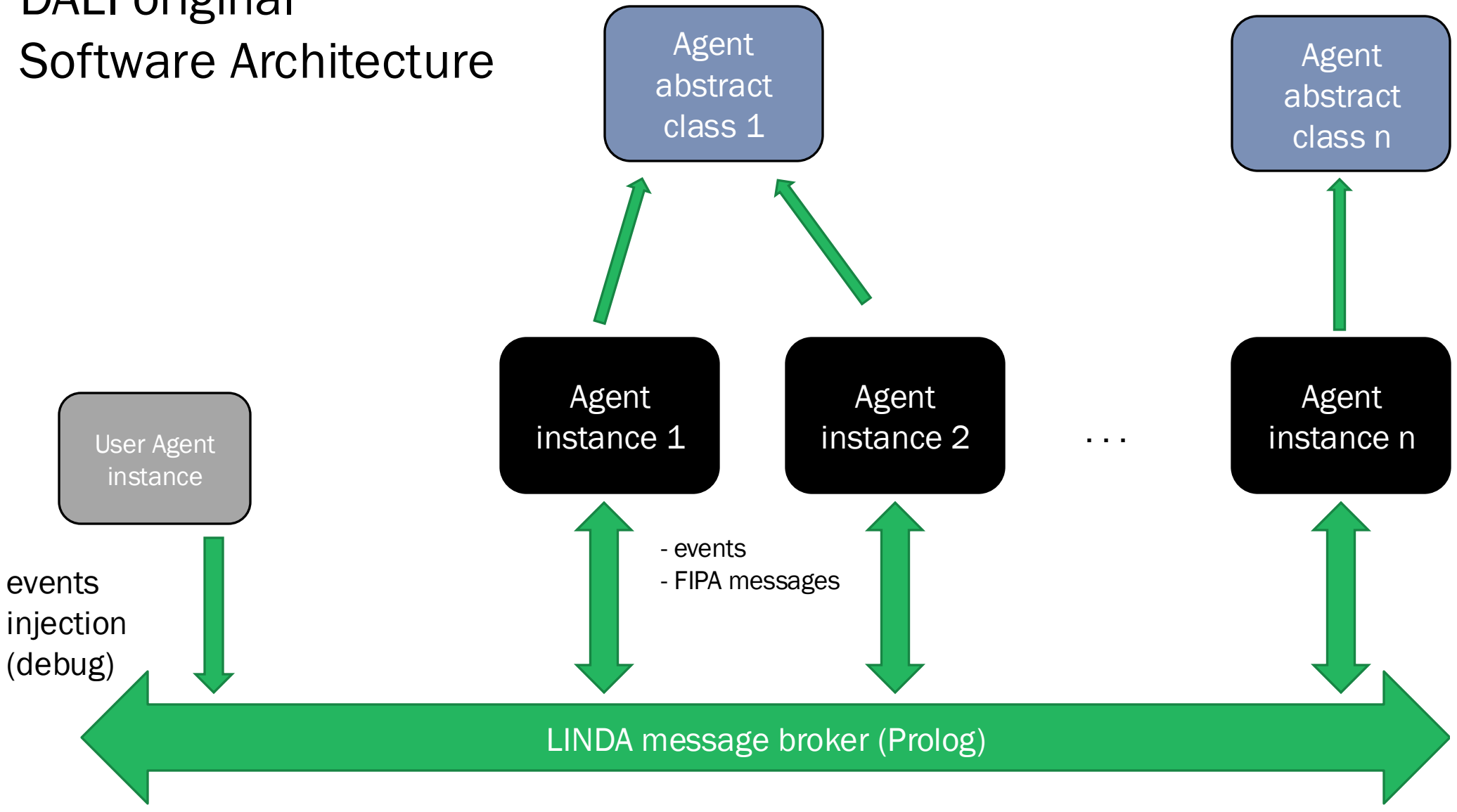
Abstract concepts	Concrete concepts
Roles	Agent Types
Permissions	Services
Responsibilities	Acquaintances
Protocols	
Activities	
Liveness properties	
Safety properties	



Learn DALI

- COSTANTINI, Stefania. *The DALI Agent-Oriented Logic Programming Language: Summary and References 2015.*
- COSTANTINI S, TOCCHIO A. *A logic programming language for multi-agent systems.* Logics in Artificial Intelligence, Springer Berlin Heidelberg, 2002, pp:1-13.
- COSTANTINI, Stefania, DE GASPERIS, Giovanni, PITONI Valentina, SALUTARI Agnese. *DALI: A multi agent system framework for the web, cognitive robotic and complex event processing.*, CILC 2017, 32nd Italian Conference on Computational Logic 26-28 September 2017, Naples, Italy
- RAFANELLI, Andrea; COSTANTINI, Stefania; DE GASPERIS, Giovanni. *A Multi-Agent-System framework for flooding events. 2022*. WOA 2022: 23rd Workshop From Objects to Agents, September 1–2, Genova, Italy

DALI original Software Architecture



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DALI extended Software Architecture

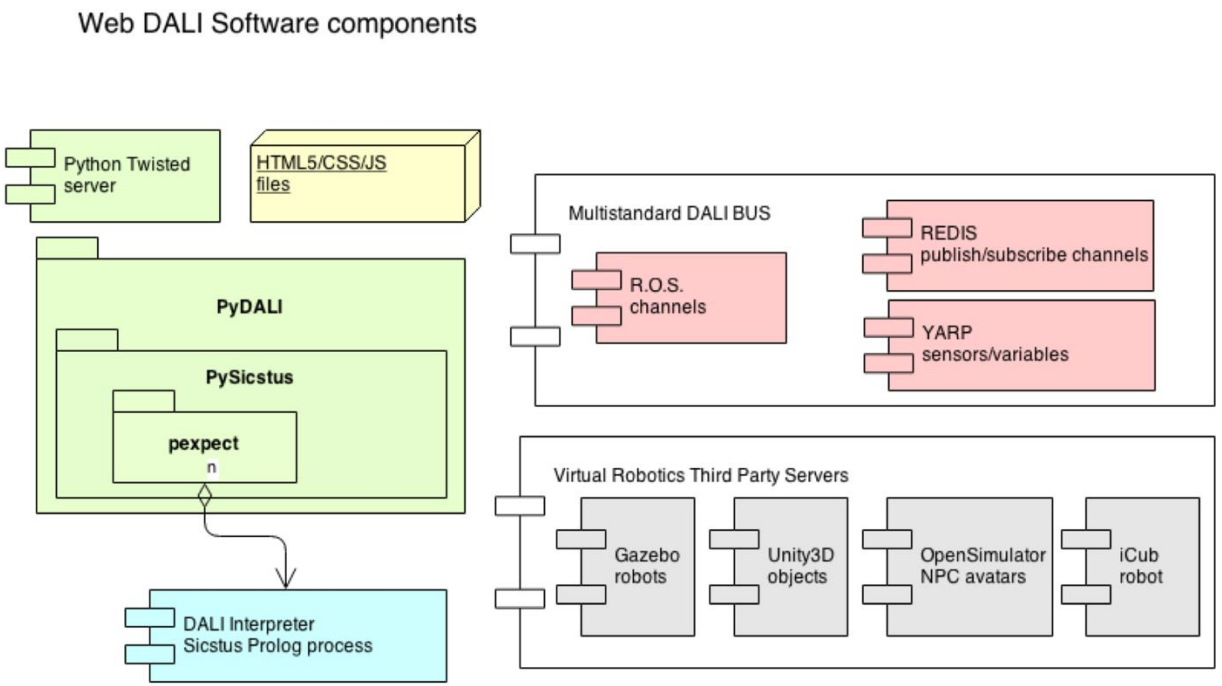

Fig. 3. Software components diagram of the extended DALI architecture

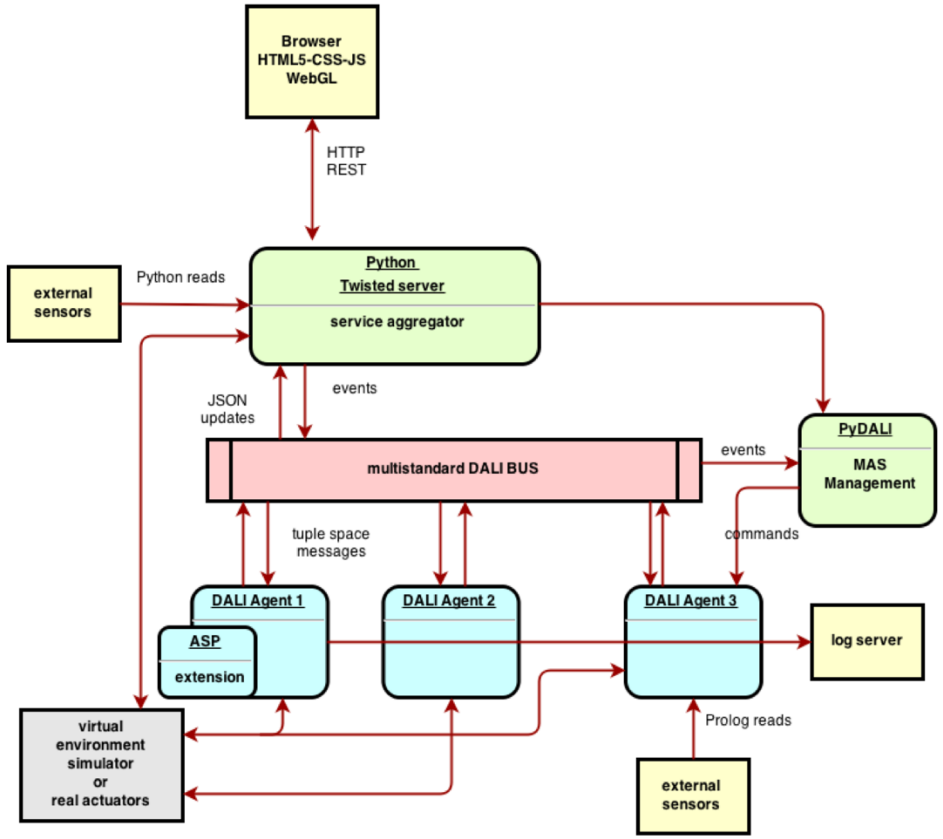

Fig. 4. Runtime deployment diagram of the extended DALI architecture

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Design your MAS

1. 0: READ THE PAPER!
2. Agent Roles
3. Inter-agent communication model with FIPA primitives
4. Virtual organisation
5. Agent classe & instances
6. Edit DALI code

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HOW-TO Install DALI

1. Install Sicstus Prolog Interpreter from the producer (V4.6.0-tested)

   https://sicstus.sics.se/download4.html

1. Clone DALI repository from GitHub
   https://github.com/AAAI-DISIM-UnivAQ/DALI

2. Test basic example Multi-Agent System (MAS)

3. Modify the basic example to try DALI syntax

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https://github.com/AAAI-DISIM-UnivAQ/DALI

AAAI-DISIM-UnivAQ / **DALI**

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DALI Multi Agent System Framework

prolog | multi-agent-systems | event-driven | logic-programming

94 commits | **2** branches | **0** packages | **13** releases | **5** contributors | Apache-2.0

Branch: **master** | New pull request | Find file | Clone or download

giodegas Merge pull request #19 from AAAI-DISIM-UnivAQ/dev ··· Latest commit `eb30e1d` on 15 Jul 2019

Examples	use Me to refer to current agent	10 months ago
docs	Added Double Events Extensions	2 years ago
src	Modified the Interpreter	2 years ago
.gitmodules	added auction examples	5 years ago
DALI_logo.png	Add files via upload	2 years ago
LICENSE	Initial commit	6 years ago
README.md	update references	2 years ago

README.md

DALI

> DALI Multi Agent Systems Framework

DALI is a meta interpreter built on top of Sicstus Prolog (R) (at the moment).

Installation

OS X & Linux:

1. To download and install SICStus Prolog (it is needed), follow the instructions at https://sicstus.sics.se/download4.html.
2. Then, you can download DALI and test it by running an example DALI MAS:

```
git clone https://github.com/AAAI-DISIM-UnivAQ/DALI.git
cd DALI/Examples/advanced
bash startmas.sh
```

You will see different windows opening:

- Prolog LINDA server (active_server_wi.pl)
- Prolog FIPA client (active_user_wi.pl)
- 1 instance of DALI metaintepreter for each agent (active_dali_wi.pl)

Windows:

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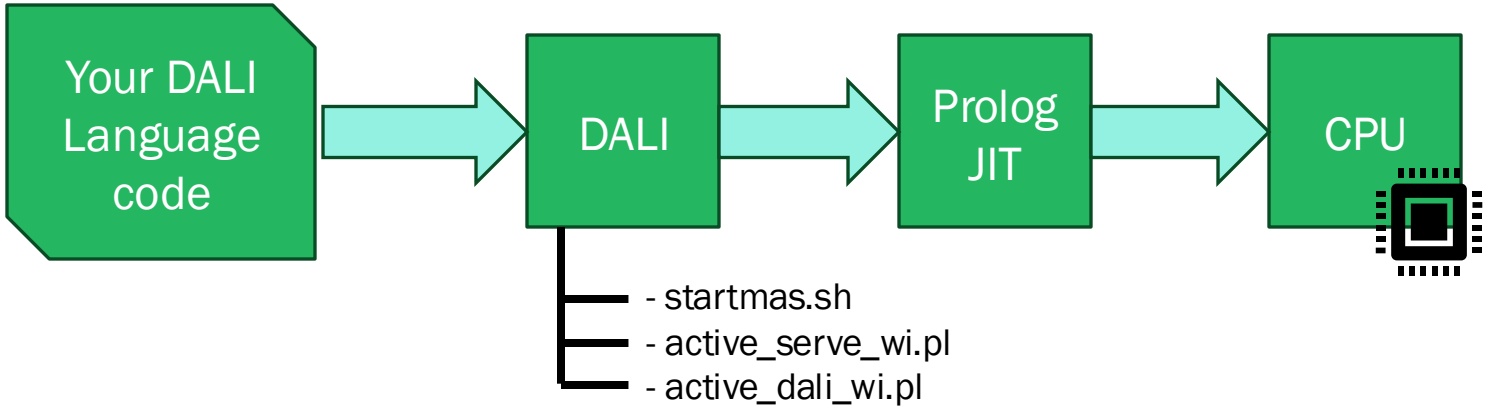
DALI Language as Prolog extension

- BNF Language definition
- Code base analysis of the meta-interpreter



DALI MAS Project requirements (minimal)

- Based on asynchronous events
- Autonomous Start and MAS coordination
- Inter-agent communication built upon different FIPA primitives (not just **`send_message`**)
- Distributed algorithm implementation
- Non-determinism

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DALI MAS Project requirements (optional-plus)

- Graphical User Interface
- 2D graphic agent representation
- 3D robot simulator interface
- Neuro-symbolic solution
- Ontology integration

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DALI Project documentation to deliver for the exam

. Sequence/collaboration diagram (UML)

. Agent types (WSL, Linux, macOS only)

. Class diagram if agent types are adopted

. Events/action table

. PDF project document **including:**

- all the above,
- install instructions tutorial in MarkDown format
- executions screenshots
- Github repository URL

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DALI Project preferred platforms

1. Docker container –> Dockerfile
2. Native Linux
3. WSL2/Linux under Windows
4. Linux VM
5. Native macOS
6. Native Windows

EDIT DALI Code

Use a programmer editor:	• Notepad++ • Sublime • Visual Studio Code with Prolog ext.
Save your agents code	• in the MAS folder
Open an advanced shell window:	• Windows Power Shell • GIT bash shell
Startup your MAS with startmas.bat (or .sh) script	• In the folder containing the MAS

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NEW ideas for A.A. 2024/25
(with plus evaluation and master degree thesis proposal)

- Integration of a LLM as source of events
- Integration of a LLM as user interface
- Refactoring the current DALI meta-interpreter code (in progress…)
- Refactoring/porting of the DALI meta-interpreter code into a open source Prolog implementation
- Develop a web-based GUI to edit-handle-start-stop the MAS
- Develop a MCP API-Interface (see MCP on Antropic web site)
- …. You propose!

Thank you for your attention, See you in the lab.

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DALI Project Development Coordinator